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Atividade 2 – Sistemas Operacionais II
Prof. Gildárcio Sousa Gonçalves

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1 – sudo apt update

sudo apt upgrade -y

```
update-initramfs: Generating /boot/initrd.img-6.8.0-106-generic
/etc/kernel/postinst.d/zz-update-grub:
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.8.0-106-generic
Found initrd image: /boot/initrd.img-6.8.0-106-generic
Found linux image: /boot/vmlinuz-6.8.0-101-generic
Found initrd image: /boot/initrd.img-6.8.0-101-generic
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-101-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-106-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...
 /etc/needrestart/restart.d/systemd-manager
 systemctl restart fwupd.service packagekit.service polkit.service rsyslog.service systemd-journald.service systemd-networkd.service systemd-resolved.service sy
stemd-timesyncd.service systemd-udevd.service udisks2.service upower.service

Service restarts being deferred:
 systemctl restart ModemManager.service
 systemctl restart systemd-logind.service

No containers need to be restarted.

User sessions running outdated binaries:
 vboxuser @ user manager service: systemd[1158]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ _
```

```
Setting up linux-tools-6.8.0-106 (6.8.0-106.106) ...
Setting up curl (8.5.0-2ubuntu10.8) ...
Setting up linux-image-generic (6.8.0-106.106) ...
Setting up linux-generic (6.8.0-106.106) ...
Setting up linux-tools-6.8.0-106-generic (6.8.0-106.106) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for linux-image-6.8.0-106-generic (6.8.0-106.106) ...
/etc/kernel/postinst.d/initramfs-tools:
update-initramfs: Generating /boot/initrd.img-6.8.0-106-generic
/etc/kernel/postinst.d/zz-update-grub:
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.8.0-106-generic
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vboxuser@Ubuntu-SO-2:~/laboratorioSOII$
```

Principais diferenças do sudo apt update para o sudo apt upgrade -y é que o sudo (super user do) apt update apenas coleta as informações das novas atualizações de softwares da Distro, não adicionando à instalação. Já o sudo apt upgrade -y atualiza tudo recebido pelo index de atualizações adquirido com o comando anterior.

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# 2 – Exploração de Comandos Básicos

pwd -> mostra o diretório atual

```
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ pwd
/home/vboxuser/laboratorioSOII
```

ls -l -> lista arquivos com detalhes

```
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ ls -l
total 0
```

cd-> navega entra diretórios

```
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ cd
vboxuser@Ubuntu-SO-2:~$ cd
```

mkdir -> cria diretórios

```
vboxuser@Ubuntu-SO-2:~$ mkdir teste
vboxuser@Ubuntu-SO-2:~$ cd teste
vboxuser@Ubuntu-SO-2:~/teste$
```

rm -r -> remove diretórios

```
vboxuser@Ubuntu-SO-2:~$ rm -r teste
vboxuser@Ubuntu-SO-2:~$
```

cp -> copia arquivos. echo “ ” > -> cria um arquivo já com caracteres

```
vboxuser@Ubuntu-SO-2:~$ echo ~arquivo teste~ > teste.txt
vboxuser@Ubuntu-SO-2:~$ ls
laboratorioSOII  teste.txt
vboxuser@Ubuntu-SO-2:~$ cp teste.txt
cp: missing destination file operand after 'teste.txt'
Try 'cp --help' for more information.
vboxuser@Ubuntu-SO-2:~$ cd teste
-bash: cd: teste: No such file or directory
vboxuser@Ubuntu-SO-2:~$ mkdir teste
vboxuser@Ubuntu-SO-2:~$ cp teste.txt teste
vboxuser@Ubuntu-SO-2:~$ cd teste/
vboxuser@Ubuntu-SO-2:~/teste$ ls
teste.txt
vboxuser@Ubuntu-SO-2:~/teste$
```

mv -> move/renomeia arquivos

```
vboxuser@Ubuntu-SO-2:~/teste$ mv teste
mv: missing destination file operand after 'teste'
Try 'mv --help' for more information.
vboxuser@Ubuntu-SO-2:~/teste$ mv teste.txt texte2.txt
vboxuser@Ubuntu-SO-2:~/teste$ ls
texte2.txt
vboxuser@Ubuntu-SO-2:~/teste$
```

cat -> concatena dados de diferentes arquivos, pode ser combinado com um navegador como vim ou nano. more ou less

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```
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ echo ~arq1 ~ arq1.txt
~arq1 /home/vboxuser arq1.txt
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ echo ~arq1 ~ > arq1.txt
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ echo ~arq2 ~ > arq2.txt
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ cat arq1.txt arq2.txt
~arq1 /home/vboxuser
~arq2 /home/vboxuser
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$
```

less -> navega em arquivos de texto

```
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ less arq1.txt
~arq1 /home/vboxuser
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$
```

whoami -> who am I, quem sou eu. Descobre o usuário atual

```
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ whoami
vboxuser
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$
```

uname -a -> exibe informações do Sistema

```
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$ uname -a
Linux Ubuntu-SO-2 6.8.0-101-generic #101-Ubuntu SMP PREEMPT_DYNAMIC Mon Feb  9 10:15:05 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
vboxuser@Ubuntu-SO-2:~/laboratorioSOII$
```

# 3 – Gerenciamento de Pacotes

vim – install/delete com sudo

```
+cmdline_info      +libcall           +python3           +virtualedit
+comments          +linebreak         +quickfix          +visual
+conceal           +lispindent        +reltime           +visualextra
+cryptv            +listcmds          +rightleft         +vreplace
+cscope            +localmap          -ruby              +wildignore
+cursorbind        -lua               +scrollbind        +wildmenu
+cursorshape       +menu              +signs             +windows
+dialog_con        +mksession         +smartindent       +writebackup
+diff              +modify_fname      +sodium            -X11
+digraphs          +mouse             -sound             +xattr
-dnd               -mouseshape        +spell             -xfontset
-ebcdic            +mouse_dec         +startuptime       -xim
+emacs_tags        +mouse_gpm         +statusline        -xpm
+eval              -mouse_jsbterm     -sun_workshop      -xsmp
+ex_extra          +mouse_netterm     +syntax            -xterm_clipboard
+extra_search      +mouse_sgr         +tag_binary        -xterm_save
-farsi             -mouse_sysmouse    -tag_old_static
   system vimrc file: "/etc/vim/vimrc"
     user vimrc file: "$HOME/.vimrc"
 2nd user vimrc file: "~/.vim/vimrc"
      user exrc file: "$HOME/.exrc"
       defaults file: "$VIMRUNTIME/defaults.vim"
  fall-back for $VIM: "/usr/share/vim"
Compilation: gcc -c -I. -Iproto -DHAVE_CONFIG_H -Wdate-time -g -O2 -fno-omit-frame-pointer -mno-omit-leaf-frame-pointer -ffile-prefix-map=/build/vim-Dkjv0L/vim-
9.1.0016=. -flto=auto -ffat-lto-objects -fstack-protector-strong -fstack-clash-protection -Wformat -Werror=format-security -fcf-protection -fdebug-prefix-map=/b
uild/vim-Dkjv0L/vim-9.1.0016=/usr/src/vim-2:9.1.0016-1ubuntu7.10 -DSYS_VIMRC_FILE=\"/etc/vim/vimrc\" -DSYS_GVIMRC_FILE=\"/etc/vim/gvimrc\" -D_REENTRANT -U_FORTI
FY_SOURCE -D_FORTIFY_SOURCE=1
Linking: gcc -Wl,-Bsymbolic-functions -flto=auto -ffat-lto-objects -Wl,-z,relro -Wl,-z,now -Wl,--as-needed -o vim -lm -ltinfo -lselinux -lsodium -lacl -lattr -l
gpm -L/usr/lib/python3.12/config-3.12-x86_64-linux-gnu -lpython3.12 -ldl -lm
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$ sudo apt remove vim -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libsodium23 vim-runtime
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  ubuntu-server vim
0 upgraded, 0 newly installed, 2 to remove and 0 not upgraded.
After this operation, 4,248 kB disk space will be freed.
(Reading database ... 122906 files and directories currently installed.)
Removing ubuntu-server (1.539.2) ...
Removing vim (2:9.1.0016-1ubuntu7.10) ...
update-alternatives: using /usr/bin/vim.tiny to provide /usr/bin/view (view) in auto mode
update-alternatives: using /usr/bin/vim.tiny to provide /usr/bin/vi (vi) in auto mode
update-alternatives: using /usr/bin/vim.tiny to provide /usr/bin/rview (rview) in auto mode
update-alternatives: using /usr/bin/vim.tiny to provide /usr/bin/ex (ex) in auto mode
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$
```

# 4 – Gerenciamento de usuários

/etc/passwd -> lista usuários

```
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$ cat/etc/passwd
-bash: cat/etc/passwd: No such file or directory
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$ cut -d: -f1 /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
systemd-timesync
dhcpcd
messagebus
systemd-resolve
pollinate
polkitd
syslog
uuidd
tcpdump
tss
landscape
fwupd-refresh
usbmux
vboxuser
aluno
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$
```

# 5 – Manipulação de Arquivos e Permissões

chmod 744 teste.txt > muda permissões

sudo chown aluno:aluno teste.txt -> altera o proprietário do arquivo

```
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$ echo "Manipulacao de arquivos e permissoes" > teste.txt
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$ ls -l
total 4
-rw-rw-r-- 1 vboxuser vboxuser 37 Mar 17 16:18 teste.txt
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$ ls
teste.txt
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$ ls -l teste.txt
-rw-rw-r-- 1 vboxuser vboxuser 37 Mar 17 16:18 teste.txt
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$ chmod 744 teste.txt
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$ sudo chown aluno:aluno teste.txt
[sudo] password for vboxuser:
vboxuser@Ubuntu-SO-2:~/laboratorioSOII/lab2$
```

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# Reflexão sobre o aprendizado:

Consegui reter conhecimentos sobre comandos básicos como mkdir(criar diretório), vim/nano/more/less(editores e visualizadores), comandos de navegação como cd/ e cd.., comandos de usuário e permissões com sudo(super user do), listagem com ls, concatenação com com e o mais importante de todos... finalmente eu sei para o que serve sudo apt update e sudo apt upgrade -y, sendo usado sudo apt update para fazer um index com todas as novas atualizações e o sudo apt upgrade -y para instalar todas as atualizações desse novo index criado em todo o sistema.